

Large Language Models: An Application-Oriented Perspective

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Speaker Introduction



National University of Singapore

Master Degree, Computer Science
2003 - 2005



Nanyang Technological University Singapore

Doctor of Philosophy (Ph.D.), Computer Engineering
2007 - 2011



Research Scientist, Data Analytics

Nanyang Technological University, Intelligent Systems Centre,
Jan 2011 - Dec 2013 · 3 yrs



Research Fellow, Data Analytics

National University of Singapore, Tropical Marine Science Institute
Jan 2014 - Oct 2014 · 10 mos



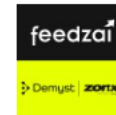
Technologist III, Data Scientist

AkzoNobel, Research Development and Innovation, Marine Coating SMU
Oct 2014 - Jul 2015 · 10 mos



Data Scientist

NEC Asia Pacific Pte Ltd
Jul 2015 - Oct 2016 · 1 yr 4 mos



Data Scientist

DemystData
Oct 2016 - Sep 2019 · 3 yrs



Senior Data Scientist

Sopra Steria



Principal Data Scientist

Singapore Institute of Manufacturing Technology



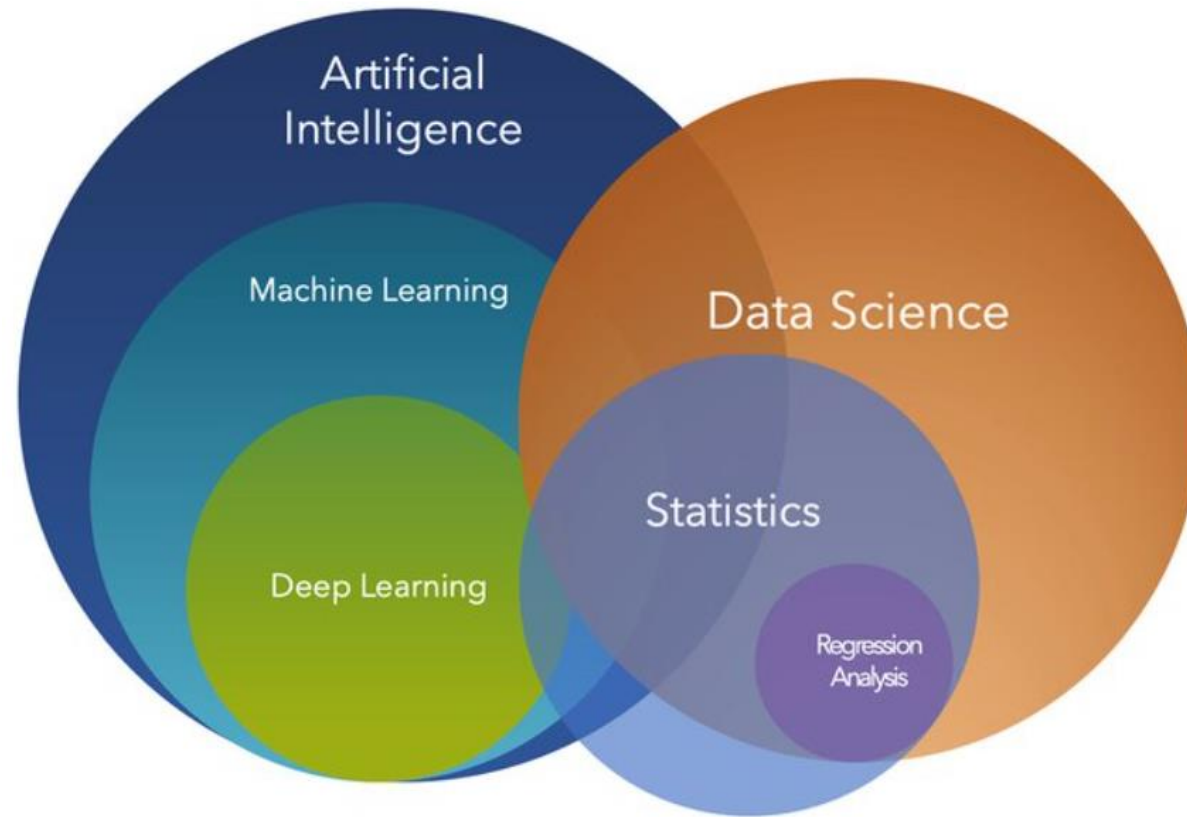
Research Fellow in Data Analytics

University of Technology Sydney
Dec 2023 - Present · 1 yr 6 mos

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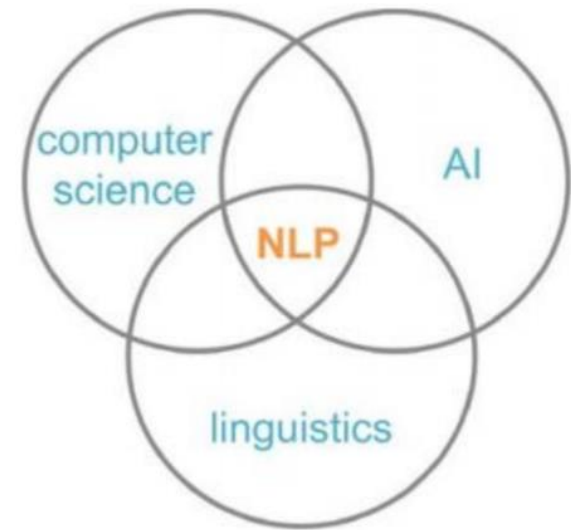
Relationships AI/ML/DS: Venn Diagram

Venn Diagram: AI,
ML, Data Science
Overlap



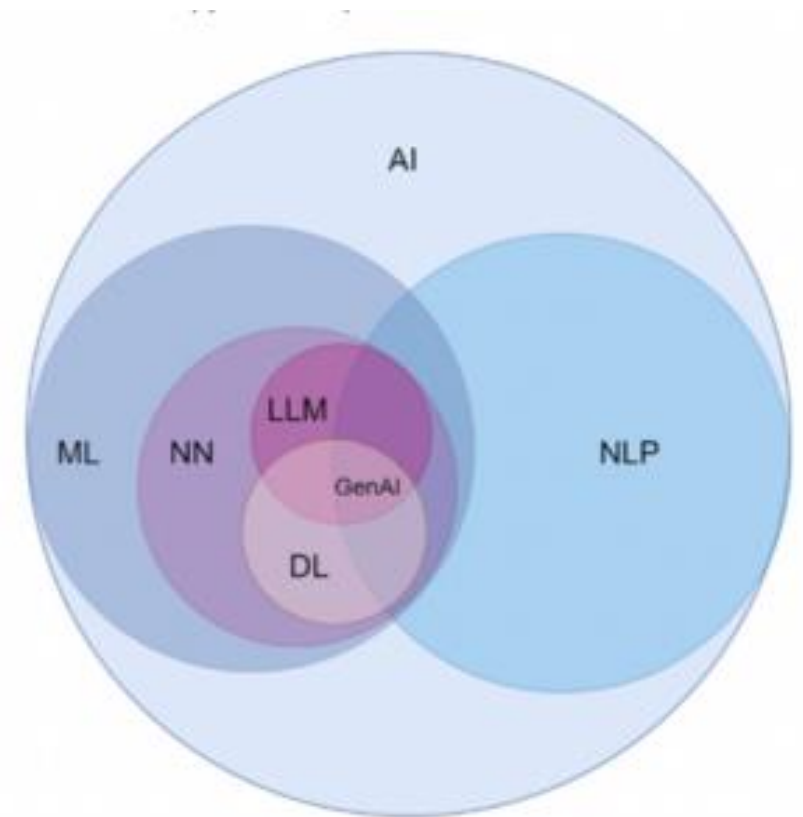
What is NLP

- Definition: the intersection of linguistics, computer science, and AI focused on enabling machines to understand human language
- Goals: language understanding (parsing meaning), generation (producing text), and translation
- Challenges: ambiguity, variability (dialects, slang), context dependency
- Real-world uses: chatbots, search engines, voice assistants
- High-level NLP pipeline: data collection → preprocessing → modeling → evaluation



NLP vs LLMs

Feature	Natural Language Processing (NLP)	Large Language Models (LLMs)
Scope	Broader field encompassing many techniques	A specific type of model within NLP
Focus	Understanding and processing language at various levels	Generating and understanding human-like text at scale
Underlying Tech	Varies (rule-based, statistical, machine learning)	Primarily deep learning, especially transformer architectures
Data Needs	Can work with smaller, task-specific datasets	Requires massive datasets for training
Capabilities	Task-specific (e.g., sentiment, NER)	Broader range, including generation, translation, reasoning



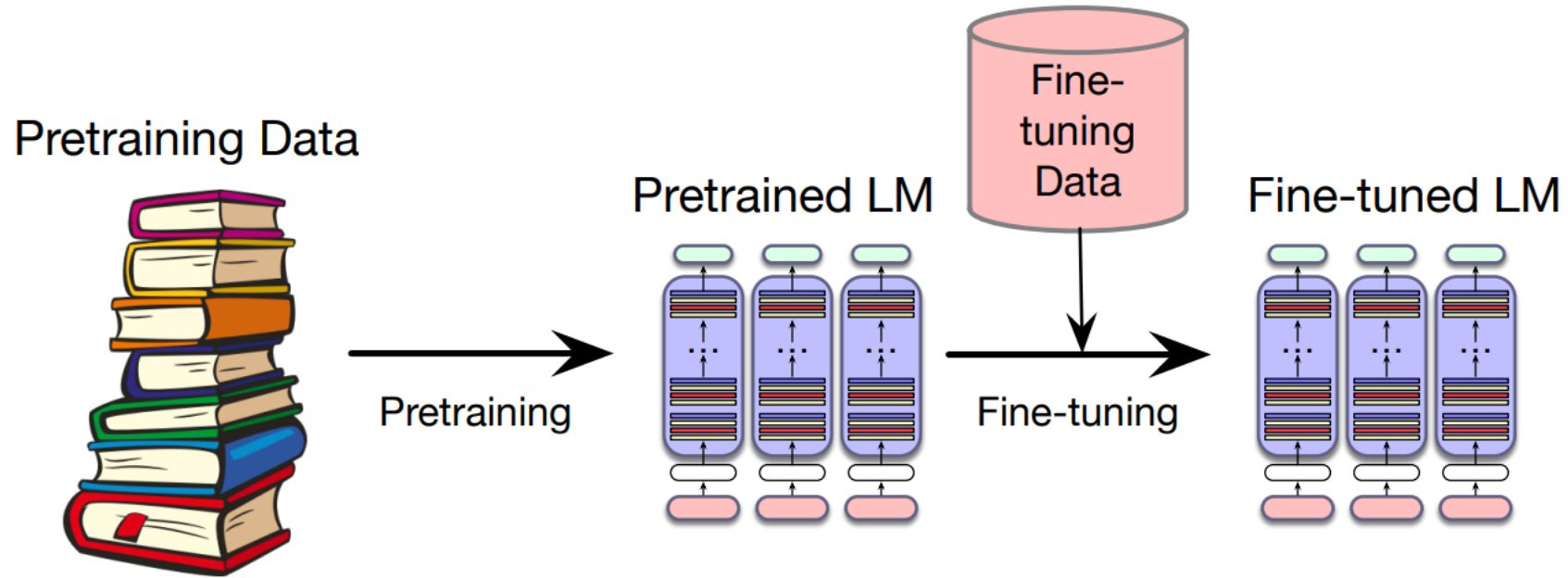
Finetuning for adaptation to new domains

What happens if we need our LLM to work well on a domain it didn't see in pretraining?

Perhaps some specific medical or legal domain?

Or maybe a multilingual LM needs to see more data on some language that was rare in pretraining?

Finetuning



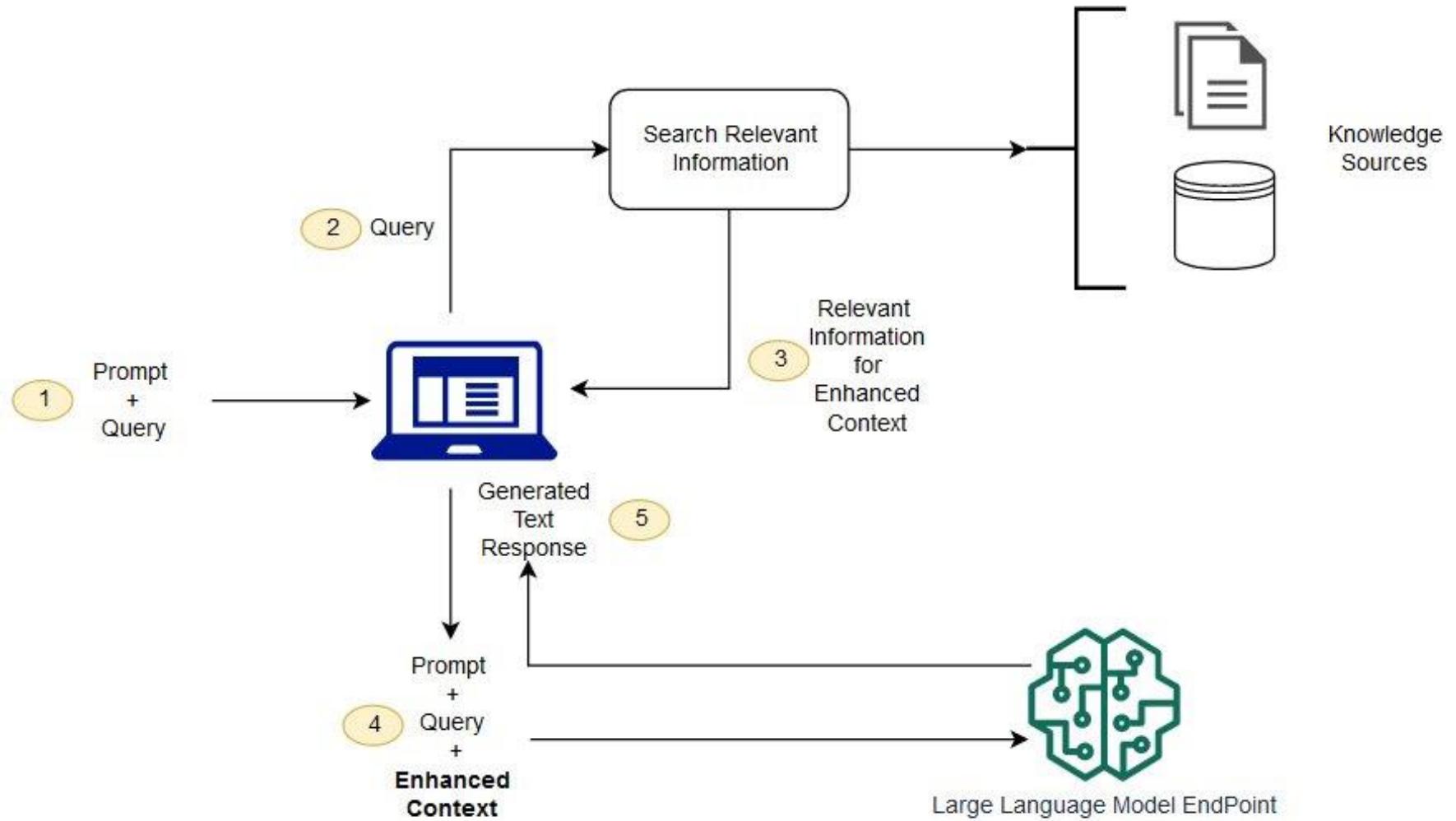
What is Retrieval-Augmented Generation?

Retrieval-Augmented Generation (RAG) is the process of optimizing the output of a large language model, so it references an authoritative knowledge base outside of its training data sources before generating a response.

Large Language Models (LLMs) are trained on vast volumes of data and use billions of parameters to generate original output for tasks like answering questions, translating languages, and completing sentences.

RAG extends the already powerful capabilities of LLMs to specific domains or an organization's internal knowledge base, all without the need to retrain the model. It is a cost-effective approach to improving LLM output so it remains relevant, accurate, and useful in various contexts.

RAG



RAG vs Fine-tuning

- **Fine-tuning:**

- This involves training a pre-trained LLM on a specific dataset to make it better at a particular task or domain. The model's parameters are updated during this process.

- **RAG:**

- This approach uses a pre-trained LLM, but it provides the model with access to external knowledge sources (like a database or knowledge graph) during **inference**. The LLM then uses this external knowledge to generate more accurate and relevant responses.

RAG vs Fine-tuning

- **Key Differences:**

- **Training:** Fine-tuning involves training the model, while RAG doesn't require model retraining.
- **Data:** Fine-tuning uses a specialized dataset to refine the model's knowledge, while RAG accesses external knowledge sources dynamically during inference.
- **Purpose:** Fine-tuning is used to improve the model's performance on specific tasks, while RAG is used to make the model more knowledgeable and up-to-date.

RAG and fine-tuning are distinct techniques for customizing LLMs, each with its own strengths and weaknesses. RAG focuses on providing the model with access to external knowledge, while fine-tuning focuses on adjusting the model's parameters through further training.

Prompt Engineering

Prompt engineering is the art and science of designing and optimizing prompts (inputs) for AI models, particularly large language models (LLMs), to achieve desired outputs.

It involves carefully crafting prompts to guide the AI toward generating specific, meaningful, and relevant responses.

Essentially, it's about providing the AI with instructions, context, and examples to help it understand your intent and produce the desired output.

Our team's research in LLMs

Limitation of hardware resource

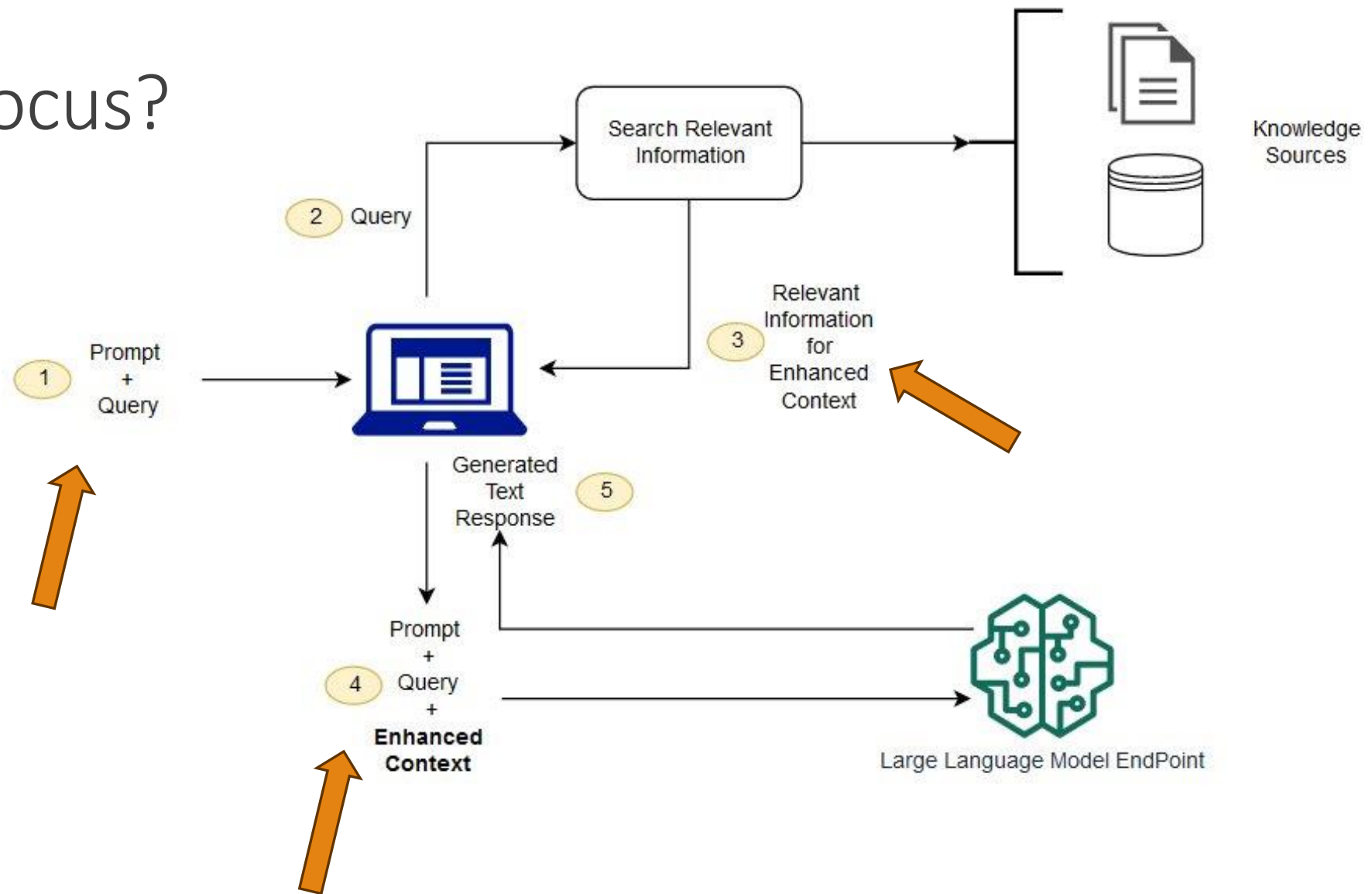
RAG is the current approach for research

Open source: Ollama

Much of our work are data collection and building the database for LLMs or Multimodals

What to focus?

RAG



Text-to-SQL

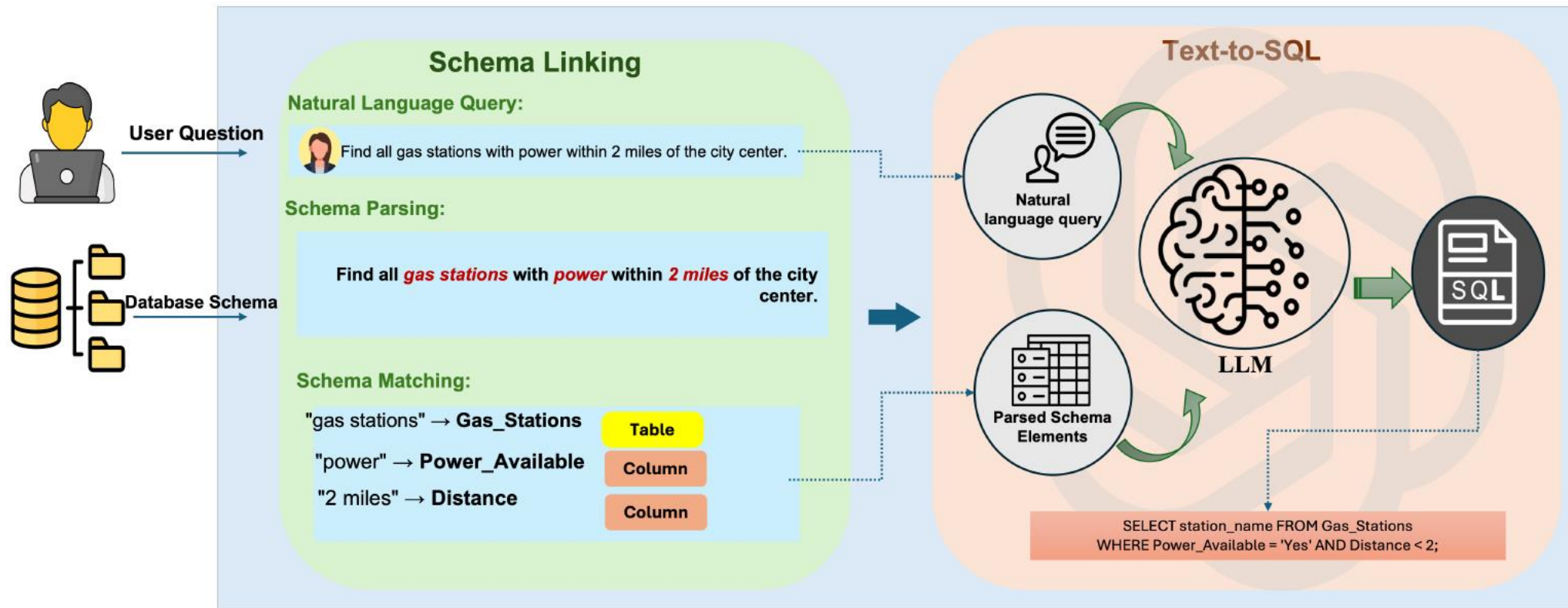


Table-to-Text

Title: 1964 United States Presidential Election in Illinois

Party	Candidates	Votes	Votes %
Democratic	Lyndon B. Johnson	2,796,833	59.47%
Republican	Barry Goldwater	1,905,946	40.53%
(...abbreviation...)			

Information Seeking Scenario 1: Data Insight Generation



Here are some meaningful insights from the given table about the 1964 United States presidential election in Illinois:

1.
2.

Information Seeking Scenario 2: Query-based Generation

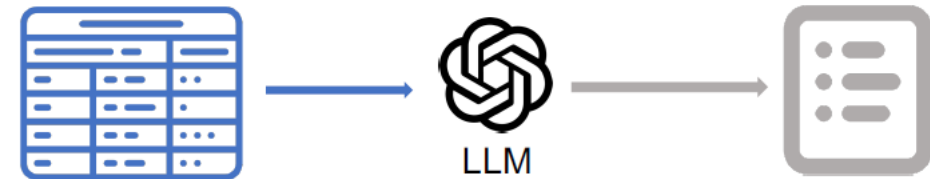


How did Lyndon B. Johnson fare against his opponent in the Illinois presidential election?

Lyndon B. Johnson won Illinois with 59.47% of the vote, against Barry Goldwater, who received 40.53% of the vote.



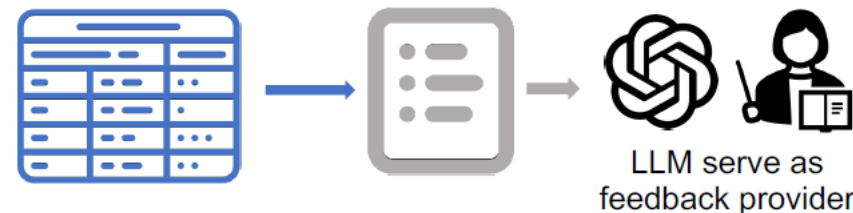
RQ1: How do LLMs perform in table-to-text generation tasks?



RQ2: Can we use LLMs to assess factual consistency of table-to-text generation?

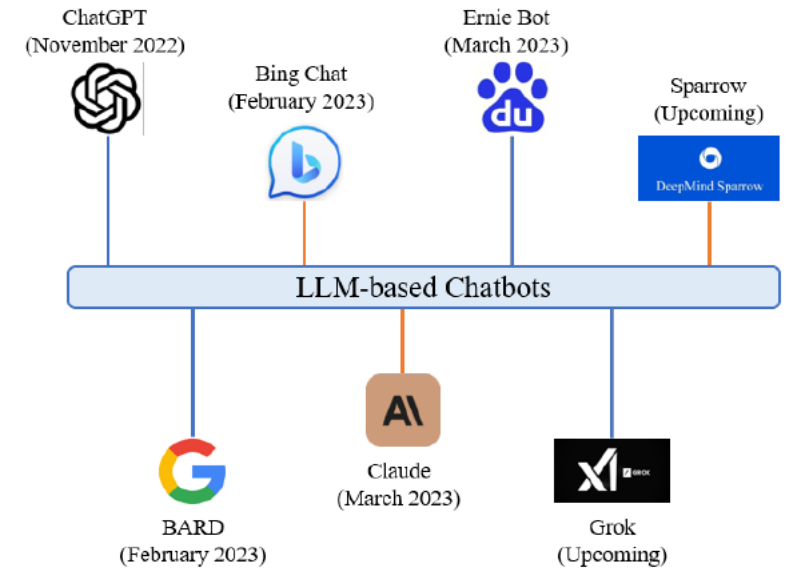
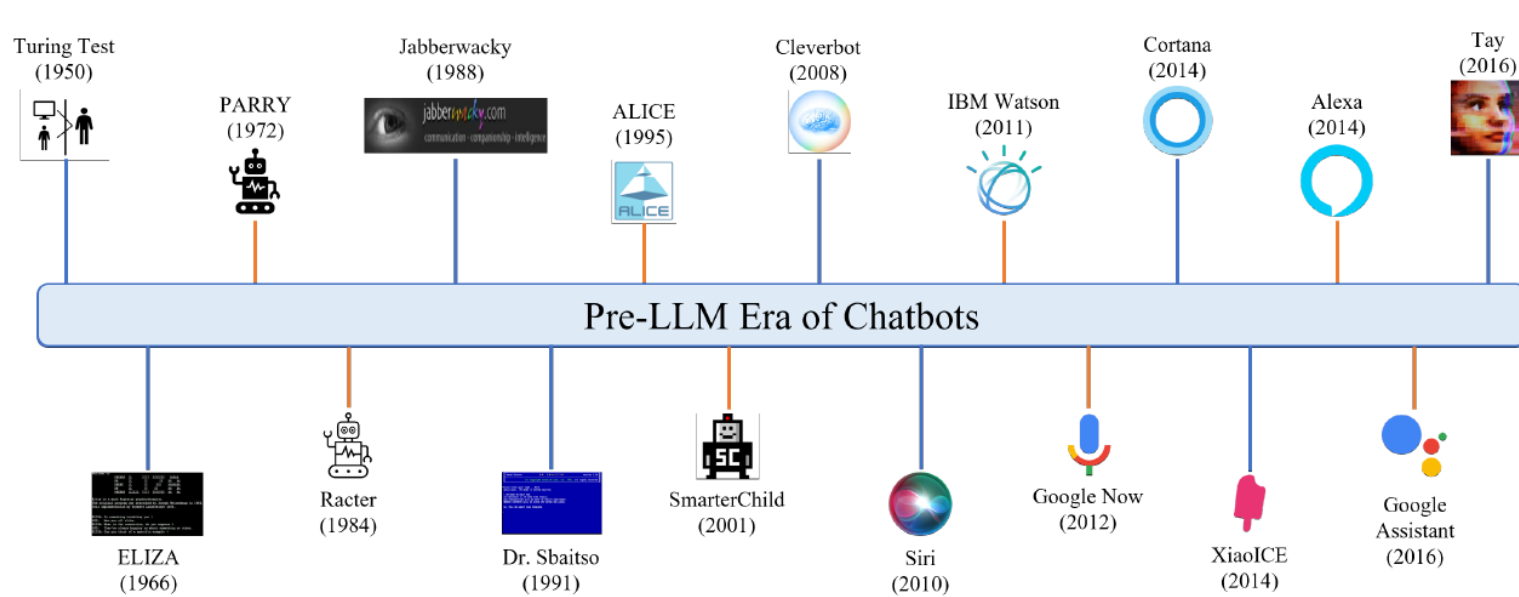


RQ3: How can fine-tuned models benefit from LLMs' strong table-to-text abilities?



LLM serve as
feedback provider

Chatbot



Timeline of LLM-based chatbots.

Use cases to apply LLMs

Game Log Analysis

Predictive Maintenance

Property (Real estate)